Vulnerability Assessment Report

Date: January 1, 2026

System Overview

Component	Details
Hardware	High-performance CPU, 128GB RAM
Operating System	Linux (latest version)
Database	MySQL
Network	IPv4, connected to internal server network
Security	SSL/TLS encryption enabled

What We're Assessing

Focus: Access controls for the database server

Timeframe: March 2026 – May 2026 (3 months)

Framework: NIST SP 800-30 Rev. 1

Why This Matters

This database server stores customer information that employees around the world depend on every day. It directly supports sales, customer service, and business growth.

The problem: The server is publicly accessible, which creates serious risks to data security.

If this server goes down or gets breached, the company faces:

- Disrupted operations
- Financial losses
- Damage to reputation
- Potential regulatory fines

Risk Assessment Summary

I identified three main threats based on likelihood and business impact:

Threat	What Could Happen	Likelihood (1–3)	Severity (1–3)	Risk Score
External Hacker	Steals customer data	3 (High)	3 (High)	**9**
Insider Misuse	Employee alters or leaks records	2 (Medium)	3 (High)	**6**
Natural Disaster	Power outage disables server	1 (Low)	2 (Medium)	**2**

Why These Three?

1. **External hackers** are the biggest concern because the server is publicly accessible—an open invitation for attackers.
2. **Insider threats** matter because employees regularly access sensitive customer data, and not all access may be necessary.
3. **Natural disasters** are less likely but still worth planning for, especially if the server is in a coastal or disaster-prone area.

Recommended Fixes

Immediate Actions

Risk	Solution
External hackers	Remove public access; require VPN or internal network connection
Unauthorized access	Add multi-factor authentication (MFA) for all database logins
Excessive permissions	Implement role-based access—employees only get access to what they need

Ongoing Protections

Risk	Solution
Insider misuse	Enable logging and regular audits of all database queries
Data theft	Encrypt sensitive data at rest, not just in transit
Natural disasters	Set up automated backups and a disaster recovery plan

Summary

The current setup leaves customer data exposed to external attacks and insider misuse. By restricting access, adding authentication layers, and preparing for outages, the company can protect its most critical asset—customer trust—while maintaining business continuity.